9.B Identify Horizontal, Vertical, and Slant Asymptotes

Rational functions are defined by the relationship between two polynomials. Their behavior is often determined not by individual points, but by how the function behaves near values where it becomes undefined or grows without bound.

Asymptotes describe this behavior. By analyzing the structure of a rational function, we can determine vertical, horizontal, and slant asymptotes without graphing, allowing us to predict how the function behaves across its domain.

"Asymptotes reveal behavior where the function cannot exist."

Learning Targets

I can determine vertical asymptotes from the denominator.
I can determine horizontal asymptotes using degree analysis.
I can determine slant asymptotes using division or structure.
I can explain how asymptotes influence graph behavior.

Introduction

When a rational function is undefined or grows very large, its graph approaches specific lines called asymptotes. These lines do not represent values the function reaches, but they describe how the function behaves.
By analyzing the structure of a rational function, we can determine these asymptotes without graphing. This allows us to predict the overall behavior of the function before sketching it.

Key Idea: Asymptotes are determined by the structure of the function, not by plotting points.

Connections

Prior Knowledge:
- Factoring polynomials to identify zeros and restrictions
- Understanding function notation and domain
- Graphing transformations of functions (from 9.A)

Cross-Disciplinary Connections:
- Physics: quantities approaching limits (e.g., intensity, force)
- Chemistry: concentration and reaction rate relationships
- Computer Science: asymptotic behavior in algorithm efficiency

Real-World Applications:
- Situations where values grow large or approach boundaries
- Modeling systems with restrictions or undefined values
- Understanding behavior near limits in applied contexts

History

The concept of asymptotes emerged as mathematicians began studying functions that did not behave like simple curves. In the 17th century, René Descartes and others developed coordinate geometry, allowing equations to be visualized as graphs.
As mathematicians explored rational functions, they noticed that graphs approached certain lines without ever touching them. These lines became known as asymptotes, providing insight into the long-term behavior of functions.
Today, asymptotes are essential in calculus and applied mathematics, where understanding limits and behavior near undefined values is critical.

Key Concepts

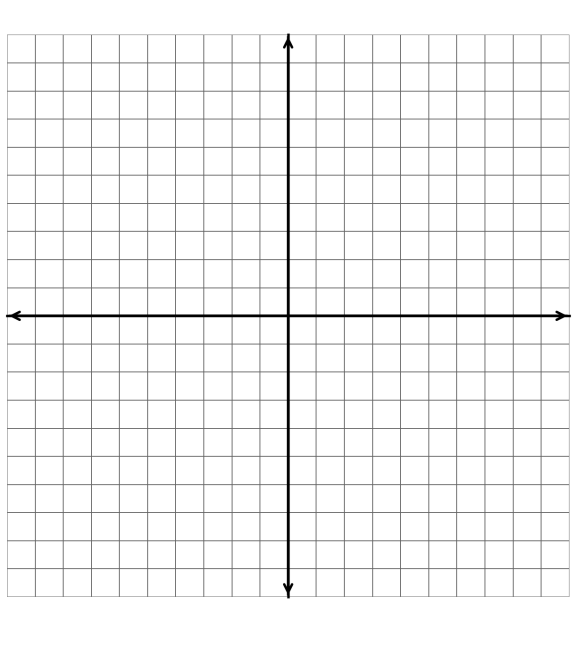
Vertical Asymptotes

Vertical asymptotes occur where the denominator of a rational function equals zero, provided the factor does not cancel.

Process:
- Factor the denominator
- Set each factor equal to zero
- Exclude any factors that cancel with the numerator

Example:

$$f(x) = \frac{1}{x-3} \quad \Rightarrow \quad \text{VA: } x = 3$$



Horizontal Asymptotes

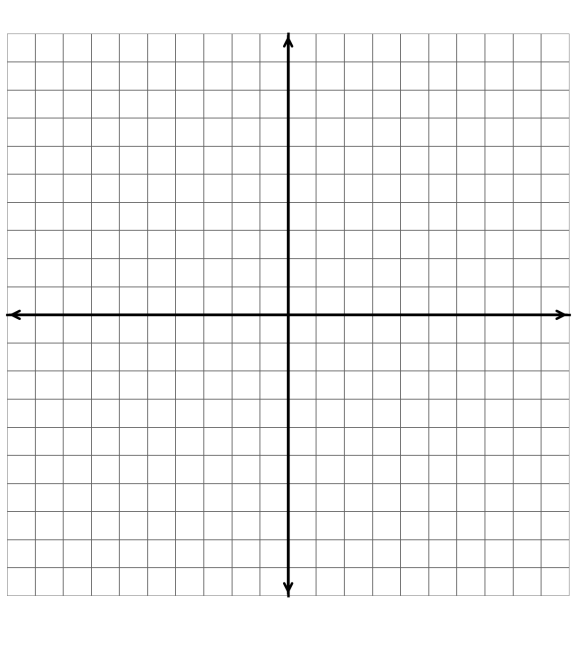
Horizontal asymptotes are determined by comparing the degrees of the numerator and denominator.

Degree Rules:

Case	Result
Degree (top) < Degree (bottom)	$y = 0$
Degree (top) = Degree (bottom)	Ratio of leading coefficients
Degree (top) > Degree (bottom)	No horizontal asymptote

Example:

$$f(x) = \frac{2x^2}{x^2+1} \quad \Rightarrow \quad \text{HA: } y = 2$$



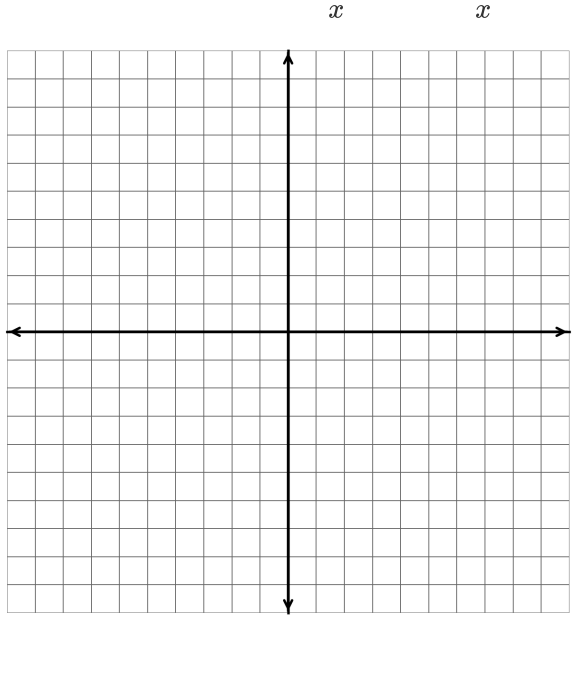
Slant (Oblique) Asymptotes

Slant asymptotes occur when the degree of the numerator is exactly one higher than the degree of the denominator.

Process:
- Use polynomial division
- The quotient (without remainder) is the slant asymptote

Example:

$$\frac{x^2+1}{x} = x + \frac{1}{x} \quad \Rightarrow \quad \text{SA: } y = x$$



Structural Behavior

Asymptotes define how a rational function behaves near undefined values and as x becomes very large or very small. The graph approaches these lines but does not cross vertical asymptotes.

Key Idea: Always determine asymptotes before graphing. They define the structure of the function.

Vocabulary

Rational Function:
A function written as the ratio of two polynomials.

Vertical Asymptote:
A vertical line $x = a$ where the function is undefined and the graph approaches but does not cross.

Horizontal Asymptote:
A horizontal line $y = b$ that the graph approaches as x becomes very large or very small.

Slant (Oblique) Asymptote:
A diagonal line that the graph approaches when the degree of the numerator is exactly one greater than the degree of the denominator.

Degree:
The highest power of the variable in a polynomial.

Leading Coefficient:
The coefficient of the term with the highest degree.

Polynomial Division:
A method used to divide polynomials to rewrite a rational function and identify slant asymptotes.

Strategies

Finding Asymptotes Efficiently:
1. **Start with the denominator.**
   Factor completely and set each factor equal to zero to identify possible vertical asymptotes.
2. **Check for cancellations.**
   If a factor cancels with the numerator, it is not a vertical asymptote (it creates a hole instead).
3. **Compare degrees.**
   Use the degrees of the numerator and denominator to determine whether a horizontal asymptote exists.
4. **Use leading coefficients.**
   When degrees are equal, divide the leading coefficients to find the horizontal asymptote.
5. **Look for slant asymptotes.**
   If the numerator's degree is exactly one higher, use division to find the slant asymptote.
6. **Use structure, not graphing.**
   Determine asymptotes algebraically before attempting to sketch the graph.

Examples

Example 1 — Vertical Asymptote

Identify the vertical asymptote.

$$f(x) = \frac{1}{x-3}$$

Example 2 — Vertical Asymptotes with Factoring

Factor and identify all vertical asymptotes.

$$f(x) = \frac{2}{x^2-5x+6}$$

Example 3 — Horizontal Asymptote (Lower Degree)

Determine the horizontal asymptote using degree analysis.

$$f(x) = \frac{3x}{x^2+1}$$

Example 4 — Horizontal Asymptote (Equal Degree)

Determine the horizontal asymptote.

$$f(x) = \frac{2x^2+1}{x^2-4}$$

Example 5 — No Horizontal Asymptote

Determine whether a horizontal asymptote exists.

$$f(x) = \frac{x^3}{x^2+1}$$

Example 6 — Slant Asymptote

Use division to determine the slant asymptote.

$$f(x) = \frac{x^2+1}{x}$$

Example 7 — Cancellation and Holes

Determine any vertical asymptotes and identify any holes.

$$f(x) = \frac{x-2}{(x-2)(x+1)}$$

Common Mistakes

- **Forgetting to factor completely:**
  Failing to factor the denominator can lead to missing vertical asymptotes.
- **Including canceled factors as asymptotes:**
  If a factor cancels, it creates a hole—not a vertical asymptote.
- **Confusing degree rules:**
  Students often mix up when the horizontal asymptote is $y = 0$ versus a ratio of coefficients.
- **Incorrect degree comparison:**
  Only the highest power matters when determining horizontal asymptotes.
- **Missing slant asymptotes:**
  If the numerator's degree is one higher, a slant asymptote exists—even though there is no horizontal asymptote.
- **Trying to graph before analyzing:**
  Asymptotes should always be determined algebraically first.

Summary

- Vertical asymptotes occur where the denominator equals zero (after simplification).
- Horizontal asymptotes are determined by comparing degrees of numerator and denominator.
- When degrees are equal, the horizontal asymptote is the ratio of leading coefficients.
- When the numerator's degree is greater by one, a slant asymptote exists.
- Structure and algebra determine asymptotes—not graphing.

Practice Set

Level 1 — Easy

1. Find the vertical asymptote.

$$f(x) = \frac{1}{x-2}$$

2. Find the vertical asymptote.

$$f(x) = \frac{3}{x+5}$$

3. Find the horizontal asymptote.

$$f(x) = \frac{2x}{x^2+1}$$

4. Find the horizontal asymptote.

$$f(x) = \frac{5}{x^3+2}$$

5. Find the vertical asymptote.

$$f(x) = \frac{4}{x-7}$$

Level 2 — Medium

6. Find all vertical asymptotes.

$$f(x) = \frac{1}{x^2-4}$$

7. Find all vertical asymptotes.

$$f(x) = \frac{2}{x^2+3x-4}$$

8. Find the horizontal asymptote.

$$f(x) = \frac{3x^2}{x^2+1}$$

9. Find the horizontal asymptote.

$$f(x) = \frac{4x^2+1}{2x^2-3}$$

10. Determine if a horizontal asymptote exists.

$$f(x) = \frac{x^3}{x^2+1}$$

Level 3 — Hard

11. Find all vertical asymptotes.

$$f(x) = \frac{x}{x^2-5x+6}$$

12. Find all vertical asymptotes.

$$f(x) = \frac{3x}{x^2+2x-3}$$

13. Find the horizontal asymptote.

$$f(x) = \frac{2x^3+1}{x^3-4}$$

14. Find the horizontal asymptote.

$$f(x) = \frac{5x^3}{2x^3+7}$$

15. Determine if a horizontal asymptote exists.

$$f(x) = \frac{x^4}{x^2+1}$$

Level 4 — Challenging

16. Find the slant asymptote.

$$f(x) = \frac{x^2+3}{x}$$

17. Find the slant asymptote.

$$f(x) = \frac{x^2-4}{x+1}$$

18. Find all asymptotes.

$$f(x) = \frac{x^2+1}{x-2}$$

19. Find all asymptotes.

$$f(x) = \frac{2x^2-3}{x+1}$$

20. Find all asymptotes.

$$f(x) = \frac{x^3}{x^2-1}$$

Level 5 — Justification / Conceptual

21. Explain why a canceled factor does not produce a vertical asymptote.

22. Explain why only the highest degree terms determine horizontal asymptotes.

23. Justify why a slant asymptote occurs when degrees differ by one.

24. Explain how asymptotes describe behavior rather than exact values.

25. Describe how asymptotes help predict graph shape without plotting points.

Answer Key

Level 1 — Easy

1. VA: $x = 2$
2. VA: $x = -5$
3. HA: $y = 0$
4. HA: $y = 0$
5. VA: $x = 7$

Level 2 — Medium

6. VA: $x = \pm 2$
7. VA: $x = 1, -4$
8. HA: $y = 3$
9. HA: $y = 2$
10. No horizontal asymptote

Level 3 — Hard

11. VA: $x = 2, 3$
12. VA: $x = 1, -3$
13. HA: $y = 2$
14. HA: $y = frac52$
15. No horizontal asymptote

Level 4 — Challenging

16. SA: $y = x$
17. SA: $y = x - 1$
18. VA: $x = 2$, SA: $y = x + 2$
19. VA: $x = -1$, SA: $y = 2x - 2$
20. VA: $x = 1, -1$, SA: $y = x$

Level 5 — Justification / Conceptual

21–25. Answers vary (conceptual explanations)

Detailed Example Solutions

Example 1 — Vertical Asymptote

$$f(x) = \frac{1}{x-3}$$

Set the denominator equal to zero:

$$x - 3 = 0 \Rightarrow x = 3$$

VA: $x = 3$

Example 2 — Factoring for Vertical Asymptotes

$$f(x) = \frac{2}{x^2-5x+6}$$

Factor the denominator:

$$x^2 - 5x + 6 = (x-2)(x-3)$$

Set each factor equal to zero:

$$x = 2, \ x = 3$$

VA: $x = 2, \ x = 3$

Example 3 — Horizontal Asymptote (Lower Degree)

$$f(x) = \frac{3x}{x^2+1}$$

Degree of numerator is less than denominator.
HA: $y = 0$

Example 4 — Horizontal Asymptote (Equal Degree)

$$f(x) = \frac{2x^2+1}{x^2-4}$$

Degrees are equal. Use leading coefficients:

$$\frac{2}{1} = 2$$

HA: $y = 2$

Example 5 — No Horizontal Asymptote

$$f(x) = \frac{x^3}{x^2+1}$$

Degree of numerator is greater than denominator.
No horizontal asymptote

Example 6 — Slant Asymptote

$$\frac{x^2+1}{x}$$

Divide:

$$\begin{aligned} x^2 + 1 &= x(x) + 1 \\ &= x + \frac{1}{x} \end{aligned}$$

SA: $y = x$

Example 7 — Cancellation and Holes

$$f(x) = \frac{x-2}{(x-2)(x+1)}$$

Cancel common factor:

$$f(x) = \frac{1}{x+1}$$

The factor $x - 2$ creates a hole, not an asymptote.
VA: $x = -1$